BioTerm-Bench: A Harbor-Compatible Benchmark for Tracking Open-Model Progress on Terminal Bioinformatics Tasks

Can an LLM agent do your routine command-line bioinformatics?

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32
self-contained terminal tasks, each graded by **comparison with a stored correct answer**

14
models evaluated on both parts of the suite, across **two independent infrastructures**

92%
of individual attempts solved by **Gemma 4 31B** — a model that fits a laptop at four bits

44%
of failures are well-formed files carrying **wrong numbers** — the mode nothing downstream catches

The question

Much of bioinformatics is not discovery. It is converting a GTF to a BED, checking that a BAM is coordinate-sorted, normalising counts to TPM — precisely specified, tedious, easy to get subtly wrong.

Which parts can be delegated to an agent, and how often is the result right?

Answering that needs a benchmark that runs the tools rather than describing them, grades deterministically rather than with an LLM judge, and runs on a laptop or an institutional node — because the data that most needs help often cannot leave the building. Human genomic data under controlled access generally cannot be sent to public generative-AI services, so the agent has to run where the data already sits.

What BioTerm-Bench is

32 deterministic tasks — a 25-task core plus a 7-task extension of multi-stage workflows. Each hands the agent real input files, a container with the usual stack, and an **exact output contract**.

Grading compares the produced artifact with a stored correct answer. **No model ever judges another model's output.**

alignment	BAM depth, duplicate marking
variant	allele frequencies, trio Mendelian
metadata	GTF → BED, longest ORF
quality control	per-base quality, FastQC summaries
search	best BLAST hit, *k*-mer Jaccard
RNA-seq	TPM normalisation, DE contrasts

Runs under **Harbor**, the Terminal-Bench harness, with the agent held fixed at **Terminus-2**, so score differences reflect the model and not the scaffolding. Four attempts per task, and every fixture is small enough to rebuild and rerun on a laptop.

The 25-task core is solved — and that is recent

Gemma 4 31B — hostable on a laptop at four bits — gets **92 %** of individual attempts right, and three models in the wider panel answer all 25 correctly on every scored attempt.

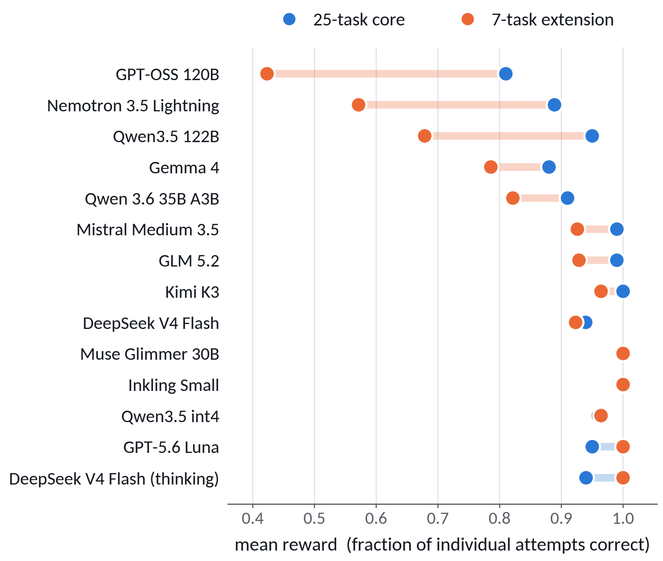
Model	mean reward	pass@1	all scored	pass@4
Gemma 4 26B A4B	0.969	25/25	22/25	25/25
Gemma 4 31B	0.920	22/25	22/25	24/25
Qwen 3.5 9B	0.878	24/25	18/25	25/25
GPT-OSS 20B	0.520	11/25	6/25	23/25
Qwen 3 14B	0.381	9/20	3/25	12/25
Mistral Small 3.2 24B	0.380	8/22	5/25	16/25
Qwen 3 30B A3B	0.283	7/24	2/25	13/25
Qwen 3 8B	0.131	3/20	1/25	6/25

OpenRouter cohort, four attempts per task, 100 trials per model. "all scored" = solved on *every* scored attempt; all denominators are explicit scored-task counts.

This says more about the moment than about the tasks. Every model in the panel was released within the last twelve months; the same 25 tasks run eighteen months ago would not have produced this table, and nine of the 32 are now failed by nobody at all.

The core keeps its value as a **continuity score** — a record of what became routinely automatable, and how fast — while the harder pack carries the discriminative load.

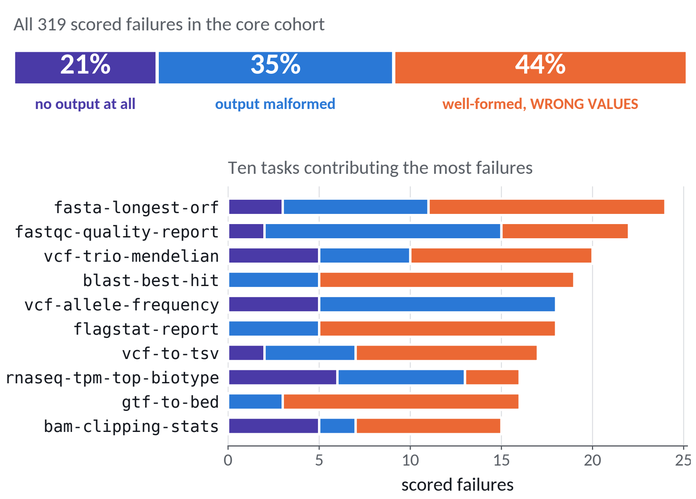
Difficulty lives in reliability, not in solvability



Every model evaluated on both parts of the suite; one row per model.

Best-of-four hides the gap. All five models of the OpenRouter cohort solve *every* extension task within four attempts, so best-of-four calls them identical. Under the **all-four** criterion — the one that matters when nobody re-runs the job — the same five span **2 of 7 to 7 of 7** — and a laboratory running each once gets a correct artifact on 100 % of attempts from one and 57 % from another.

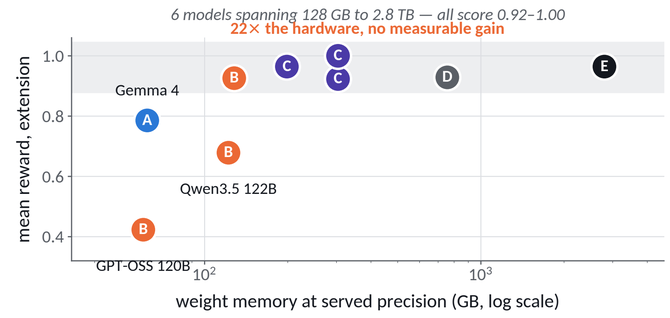
How agents fail — and why it matters



All 319 scored failures in the included core cohort, classified from the verifiers' own reasons.

The largest category is the quiet one. A BED file with correct syntax and wrong coordinates, or a flagstat summary with plausible but incorrect counts, propagates silently into whatever consumes it. The loud 56% announce themselves — a downstream tool refuses to read them. The quiet 44% do not, and they are spread across ordinary tasks rather than exotic ones.

What hardware do you actually need?

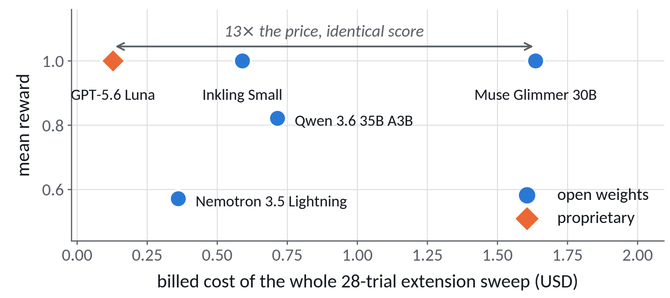


Model	Total (active)	GB	Hardware needed
GPT-OSS 120B	120B (A5.1B)	60	B: one datacentre GPU
Gemma 4	31B	62	A: laptop or workstation
Qwen3.5 122B	122B (A10B)	122	B: one datacentre GPU
Mistral Medium 3.5	128B	128	B: one datacentre GPU
Qwen3.5 int4	397B (A17B)	199	C: 2-4 GPU server
DeepSeek V4 Flash	304B (A13B)	304	C: 2-4 GPU server
DeepSeek V4 Flash Thinking	304B (A13B)	304	C: 2-4 GPU server
GLM 5.2	756B	756	D: full 8-GPU node
Kimi K3	2.8T (A104B)	2800	E: multi-node cluster

Weight memory is total parameters at the served precision — a floor that excludes KV cache and runtime overhead. Letters mark the tier; tier A assumes 4-bit quantisation.

A **62 GB workstation** model outscores a 122 GB one needing a datacentre GPU.

Price is not capability



The whole extension sweep cost **$0.13–$1.64** per model, and the cheapest run ties for the best score. Price follows parameter count and provider margin — not the licence.

Research infrastructure is the enabling layer

Nine of the fourteen models here were served by e-INFRA CZ, which offers LLM inference to **ELIXIR CZ** members on institutional hardware. That is what makes agentic analysis possible for controlled-access human data: the model runs *inside the administrative boundary* and nothing is exported.

It is also sufficient. A tier B or C model hosted this way reaches the **top of the observed score range** — no frontier API required.

What to delegate today

Delegate, with spot-checks

Format conversion and summarisation — GTF → BED, VCF → TSV, FASTA statistics, flagstat-style counting. Reliable from the strongest models — but also where weaker ones fail most, so check the artifact.

Delegate, then verify

Tasks solved most of the time and plausibly wrong otherwise. Run it twice and diff — cheap insurance against the failure mode nothing downstream will flag.

Not yet

Multi-stage reasoning over biological structure — splice-junction support, trio-aware and diploid-consensus variant logic. One such task was solved by only 2 of 14 models.

Rule of thumb: delegation is safe in proportion to how mechanically checkable the output is — a BED file can be checked by re-deriving a handful of rows, a variant set with a subtle inheritance error cannot be checked without redoing the analysis.

And report mean reward and all-attempts, not best-of-*n*. Best-of-*n* describes a workflow with an oracle telling you which run was right. You do not have one — that is why you wanted the agent.

github.com/ML-Bioinfo-CEITEC/BioTerm-Bench

Computational resources, and the served models themselves, were provided by the **e-INFRA CZ** project (ID:90254), supported by the Ministry of Education, Youth and Sports of the Czech Republic.
Tasks, images, graders, runners and every analysis script are released under Apache 2.0.